

MONETARY POLICY TRANSMISSION INTO INDIAN EQUITY MARKETS

Understanding how macroeconomics shocks impact NIFTY & BankNifty – Magnitude, timing and persistence

EXECUTIVE SUMMARY

This project analyzes how macroeconomic variables—specifically global liquidity (Fed), domestic policy (Repo rate), interest rates (US 10Y), inflation, and volatility (VIX)—impact Indian equity markets.

Using monthly data from 2007 to 2025, the analysis combines regression, lag modeling, and VAR-based time-series techniques to understand not just whether macro variables matter, but how and when they influence markets.

Key findings:

- VIX has a strong and immediate impact on both NIFTY and BankNifty (p-values < 1e-10)
- Repo rate significantly impacts BankNifty ($\beta \approx +0.057$, $p \approx 0.04$), confirming sector sensitivity
- Fed shocks show delayed but strong transmission, peaking around lag 1–2
- IRF analysis shows BankNifty reacts up to +0.093 to Fed shocks, indicating strong global linkage
- FEVD shows global factors explain ~94–97% of initial market variance

Overall, markets are driven by a mix of:

- Immediate sentiment (VIX)
- Delayed global liquidity (Fed)
- Gradual domestic policy (Repo)

BUSINESS PROBLEM

Equity markets react to macroeconomic events constantly, but the challenge is:

- Reactions are not always immediate
- Different variables impact markets differently
- Timing and persistence vary significantly

For analysts and investors, this creates uncertainty:

- Should we react immediately?
- Or wait for delayed effects?

This project solves that by answering:

- ✓ Which macro variables matter most?
- ✓ How strong is their impact?
- ✓ When does the impact occur?
- ✓ How long does it persist?

WHY THIS PROJECT

This project was chosen to bridge the gap between theoretical macroeconomics and practical market behavior.

Instead of analyzing markets in isolation, this study integrates:

- Global macro factors (Fed, US yields)
- Domestic policy (Repo rate)
- Market sentiment (VIX)

This reflects how real-world institutional analysts evaluate markets.

The goal was to build a multi-layered analytical framework that mimics how hedge funds and research desks analyze macro-driven market movements.

DATA OVERVIEW

Dataset Period: October 2007 – 2025

Frequency: Monthly

Key Variables:

Market:

- NIFTY returns
- BankNifty returns

Macro:

- Fed rate changes
- Repo rate changes
- US 10Y yield changes
- US CPI changes
- VIX changes
- US inflation (derived)

Total observations: ~210 monthly records (post-cleaning)

Data Sources:

- Yahoo Finance (market data)
- FRED API (macro variables)
- RBI policy data (repo rate timeline)

All datasets were merged into a unified master dataset and aligned by date.

DATA PREPARATION

Data preparation was performed using Python:

- Converted daily data into monthly frequency (month-end)
- Calculated returns using percentage change
- Created lag variables (L0–L3) for each macro variable
- Built a unified dataset combining:
 - Market + US macro + RBI policy

Example transformations:

- `NIFTY_Return = pct_change(NIFTY)`
- `fed_change_L1 = lagged Fed impact`
- `repo_change_L3 = delayed policy effect`

Final dataset was cleaned (NaNs removed) and used for modeling.

```
--- Monthly Data Sample ---
      NIFTY    BankNifty  NIFTY_Return  BankNifty_Return
Date
2007-10-31  5900.649902  9013.150391      0.175112      0.120732
2007-11-30  5762.750000  9375.750000     -0.023370      0.040230
2007-12-31  6138.600098  9863.450195      0.065221      0.052017
2008-01-31  5137.450195  9226.250000     -0.163091     -0.064602
2008-02-29  5223.500000  8688.349609      0.016750     -0.058301

--- Dataset Shape ---
(219, 4)

--- Column Names ---
Index(['NIFTY', 'BankNifty', 'NIFTY_Return', 'BankNifty_Return'], dtype='str')
```

```
--- Final Master Dataset Sample ---
      Date      NIFTY    BankNifty  NIFTY_Return  ...    US_10Y      VIX  US_Inflation  Repo_Rate
0 2007-10-31  5900.649902  9013.150391      0.175112  ...    4.527727  19.115652      3.610698      7.75
1 2007-11-30  5762.750000  9375.750000     -0.023370  ...    4.148500  25.582381      4.373267      7.75
2 2007-12-31  6138.600098  9863.450195      0.065221  ...    4.097500  21.651000      4.108813      7.75
3 2008-01-31  5137.450195  9226.250000     -0.163091  ...    3.744286  25.816190      4.294696      7.75
4 2008-02-29  5223.500000  8688.349609      0.016750  ...    3.737500  25.456000      4.142959      7.75

[5 rows x 11 columns]

--- Final Dataset Shape ---
(218, 11)

--- Final Columns ---
Index(['Date', 'NIFTY', 'BankNifty', 'NIFTY_Return', 'BankNifty_Return',
      'Fed_Rate', 'US_CPI', 'US_10Y', 'VIX', 'US_Inflation', 'Repo_Rate'],
      dtype='str')
```

EXPLORATORY ANALYSIS

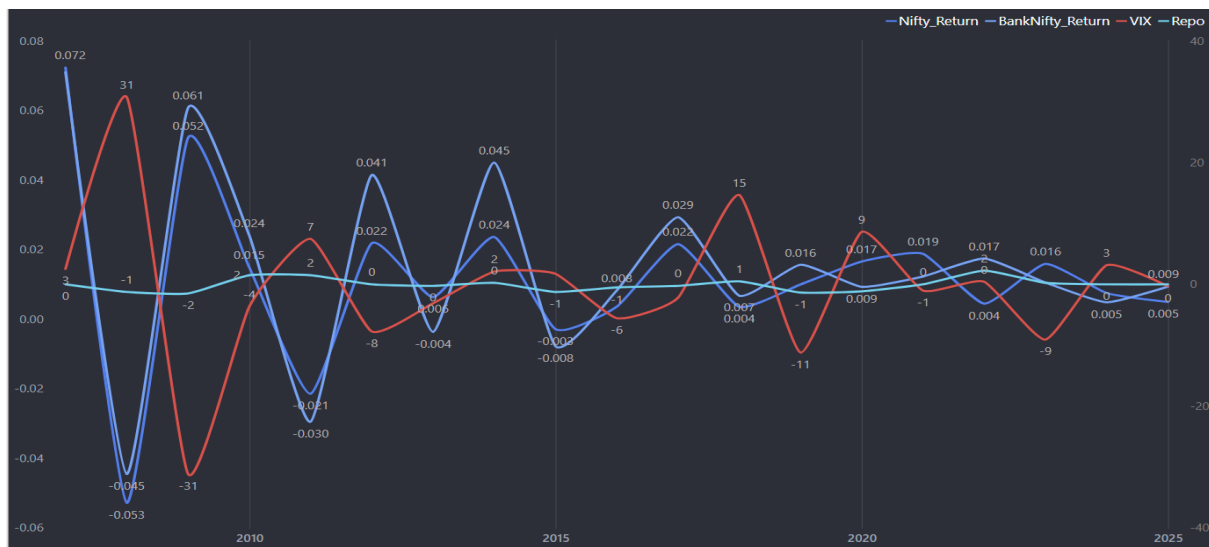
Initial statistical analysis revealed:

1. Volatility (VIX):
 - Strong negative relationship with returns
 - Highly significant:
 - NIFTY: $p \approx 1.11e-14$
 - BankNifty: $p \approx 6.19e-9$
2. Repo Rate:
 - Significant for BankNifty:
 - Coefficient $\approx +0.057$
 - $p \approx 0.040$
3. Fed Rate:
 - Not immediately significant
 - But becomes important in lag analysis
4. Other variables (US CPI, US 10Y):
 - Weak standalone significance
 - Indicate indirect or lagged effects

Conclusion:

→ Immediate drivers = VIX

→ Structural drivers = Fed & Repo



REGRESSION INSIGHTS

Regression results highlight:

- VIX consistently impacts both markets negatively
- Repo rate has sector-specific impact (BankNifty > NIFTY)
- Global variables require lag analysis to reveal true effect

Market	Variable	Coefficient	p_value	Impact
nifty_return	fed_change	0.029427537	0.176360791	Not significant
nifty_return	us_10y_change	-0.016642024	0.335844052	Not significant
nifty_return	us_cpi_change	-0.006183958	0.332983297	Not significant
nifty_return	vix_change	-0.005862199	1.11907E-14	Significant
nifty_return	us_inflation_change	-0.001348557	0.893163493	Not significant
nifty_return	repo_change	0.022177939	0.228228284	Not significant
banknifty_return	fed_change	0.051679823	0.118465951	Not significant
banknifty_return	us_cpi_change	-0.016346039	0.092943962	Not significant
banknifty_return	us_10y_change	-0.031493705	0.231039423	Not significant
banknifty_return	vix_change	-0.00649073	6.19893E-09	Significant
banknifty_return	us_inflation_change	0.002872281	0.850705701	Not significant
banknifty_return	repo_change	0.057359239	0.040963403	Significant

Example:

BankNifty vs Repo:
→ $\beta \approx +0.057$ → strong positive sensitivity

VIX vs NIFTY:
→ $\beta \approx -0.006$ → inverse relationship

Interpretation:

- Banking sector reacts strongly to policy
- Markets react instantly to volatility shocks

A negative VIX coefficient indicates that rising volatility leads to declining market returns, while a positive Repo coefficient suggests policy changes have a measurable directional effect on banking stocks.

LAG ANALYSIS (CRITICAL PART)

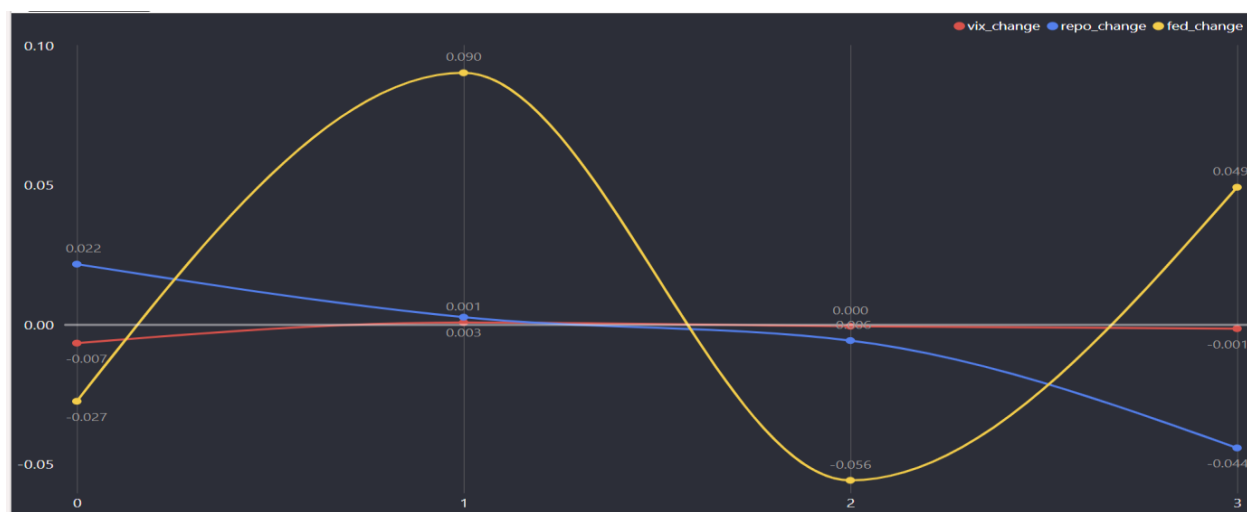
Lag analysis revealed timing differences:

- Fed shocks:
 - Peak impact at lag 1–2
 - Example: BankNifty response $\approx +0.093$ at horizon 1
- Repo shocks:
 - Gradual impact across lags
 - Example: negative impact at lag 3 ≈ -0.053
- VIX:
 - Immediate (lag 0) impact
 - Very weak persistence

Insight:

- Markets do NOT react instantly to all variables
- Timing matters as much as magnitude

Market	Variable	Coefficient	p_value	lag
nifty_return	fed_change_L0	-0.015984084	0.603825072	0
nifty_return	fed_change_L1	0.059166495	0.090151615	1
nifty_return	fed_change_L2	-0.034114631	0.32449239	2
nifty_return	fed_change_L3	0.029606499	0.316988647	3
nifty_return	vix_change_L0	-0.006019913	3.52698E-12	0
nifty_return	VIX_change_L1	0.000734863	0.372044606	1
nifty_return	VIX_change_L2	-0.000845362	0.304447701	2
nifty_return	VIX_change_L3	-0.001292183	0.084673076	3
nifty_return	repo_change_L0	0.01257062	0.526408009	0
nifty_return	Repo_change_L1	0.004171759	0.827414821	1
nifty_return	Repo_change_L2	-0.017827318	0.355941346	2
nifty_return	Repo_change_L3	-0.034623601	0.078854541	3
banknifty_return	fed_change_L0	-0.038841269	0.422357841	0
banknifty_return	fed_change_L1	0.121699934	0.026875491	1
banknifty_return	fed_change_L2	-0.077459827	0.154945984	2
banknifty_return	fed_change_L3	0.069132525	0.137481487	3
banknifty_return	vix_change_L0	-0.007083729	9.07481E-08	0
banknifty_return	VIX_change_L1	0.000969295	0.453399721	1
banknifty_return	VIX_change_L2	-0.000135601	0.91637472	2
banknifty_return	VIX_change_L3	-0.001434141	0.222292192	3
banknifty_return	repo_change_L0	0.030987755	0.32051299	0
banknifty_return	Repo_change_L1	0.001354619	0.964052179	1
banknifty_return	Repo_change_L2	0.006550805	0.82886512	2
banknifty_return	Repo_change_L3	-0.053690538	0.082743653	3



This suggests that reacting only to immediate macro changes may lead to mis-timed decisions.

IRF (DYNAMIC IMPACT)

Impulse Response analysis shows:

Fed shock:

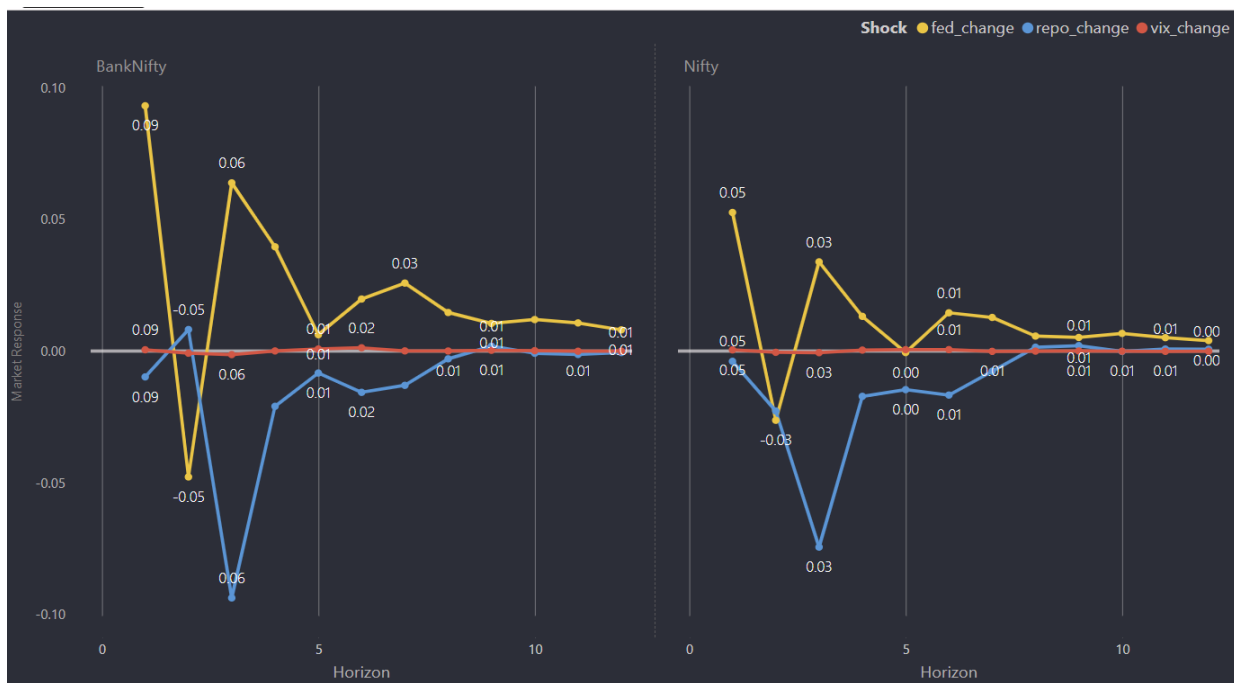
- BankNifty initial response $\approx +0.093$
- Strong but stabilizes over time

Repo shock:

- NIFTY response becomes negative (~ -0.074 at horizon 3)
- Indicates tightening impact

VIX shock:

- Immediate impact but fades quickly
- No long-term persistence



Conclusion:

- Global shocks = strong + delayed
- Policy shocks = moderate + persistent
- Sentiment shocks = immediate + short-lived

FEVD (WHO DRIVES THE MARKET?)

Variance decomposition results:

Early horizon:

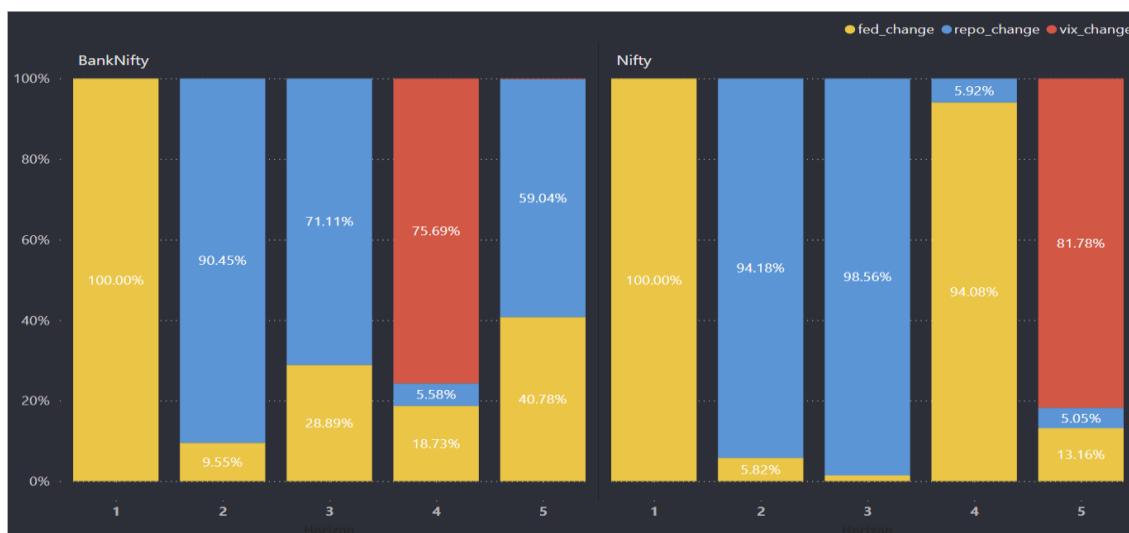
- Fed explains ~94–97% of variance
→ Dominant global influence

Mid horizon:

- Repo contribution increases (~86% at some points), indicating growing domestic influence over time.

Later horizon:

- VIX spikes (~60% in volatility periods)



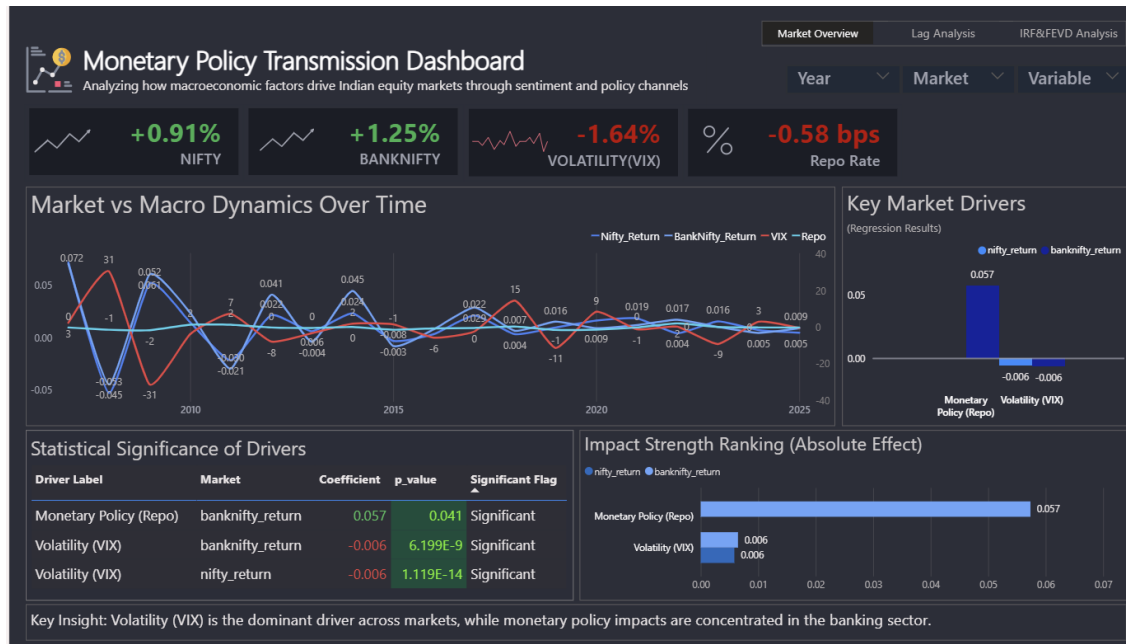
Interpretation:

- Market drivers change over time
- Early = global
- Mid = domestic
- Short-term spikes = sentiment

DASHBOARD STORY

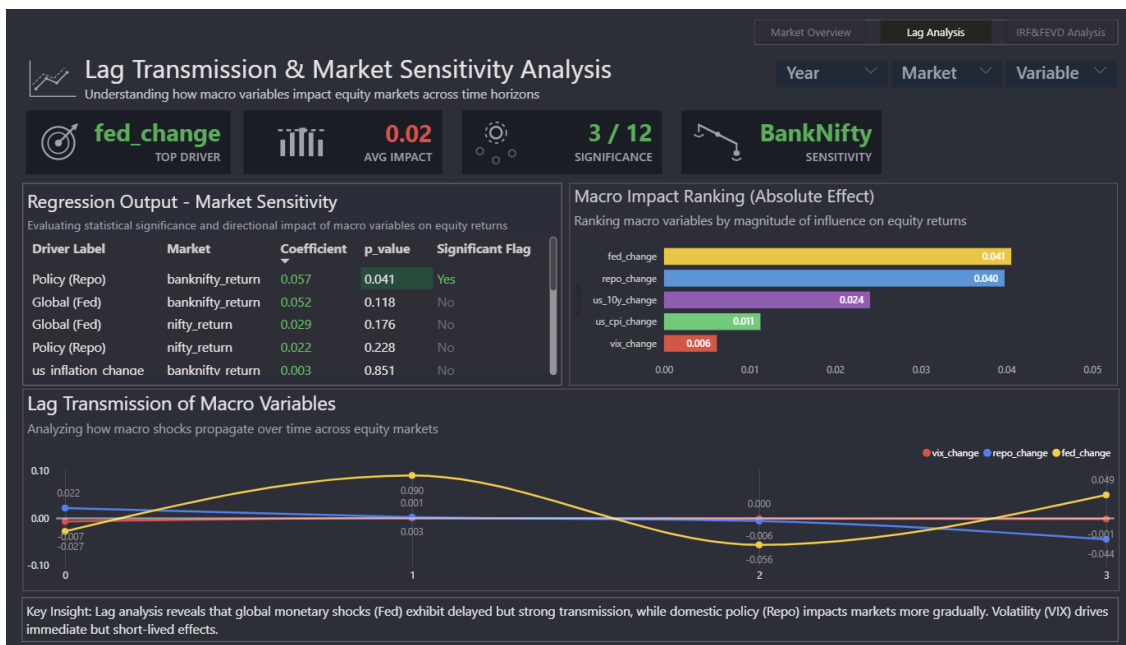
Dashboard 1:

- Market overview + key drivers
- Shows immediate relationships



Dashboard 2:

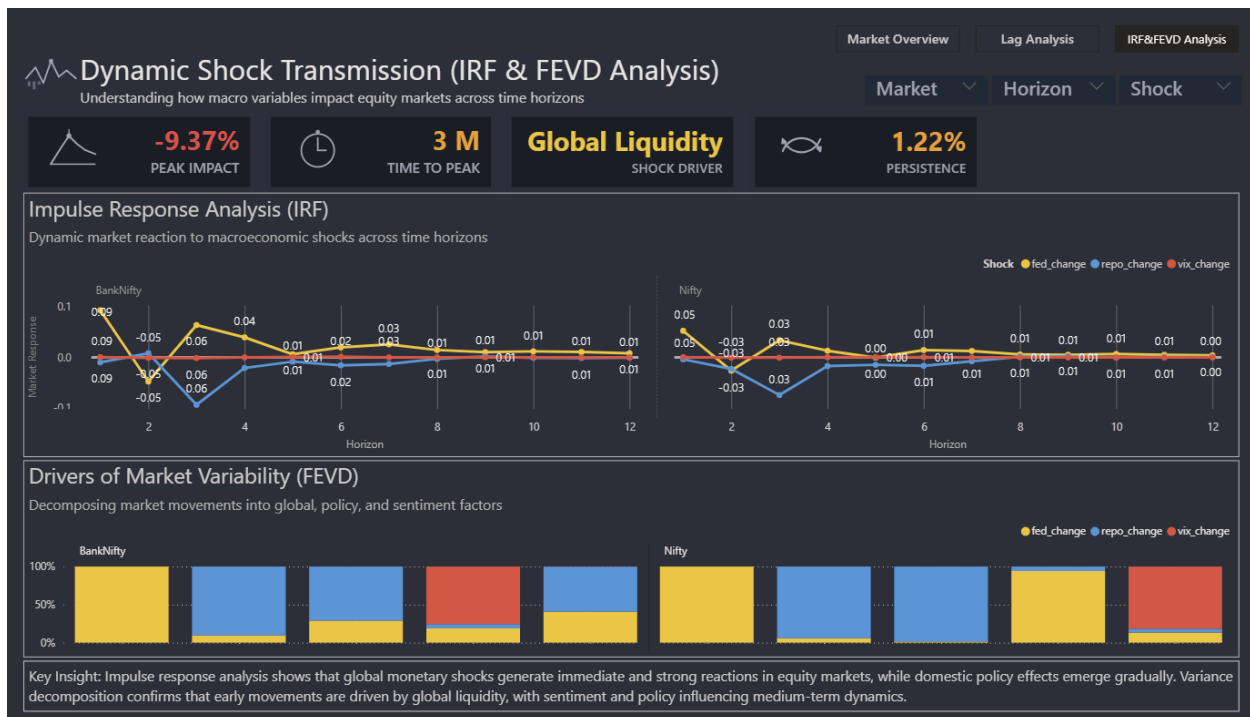
- Sensitivity + lag structure
- Explains timing of impact



Dashboard 3:

→ IRF + FEVD

→ Shows dynamic behavior and shock transmission



Together, these dashboards provide a layered analytical workflow from observation to explanation to dynamic modeling.

KEY INSIGHTS

- Volatility (VIX) is the strongest short-term driver
- BankNifty is highly sensitive to monetary policy
- Fed shocks dominate global influence
- Market reactions vary across time horizons

Timing differences across variables are as important as their magnitude

BUSINESS RECOMMENDATIONS

For portfolio managers and market analysts, the following actions can be considered:

- Use VIX for short-term trading signals
- Track Fed policy for forward positioning
- Align banking sector exposure with RBI policy
- Incorporate lag analysis into strategy
- Avoid reacting only to immediate data

FINAL CONCLUSION

This project demonstrates how macroeconomic variables influence equity markets in a structured and time-dependent manner.

By combining statistical models with dynamic analysis, it provides a realistic view of market behavior.

The framework can be extended into real-time systems for macro-driven investment strategies. This approach provides a scalable foundation for building macro-driven investment frameworks.